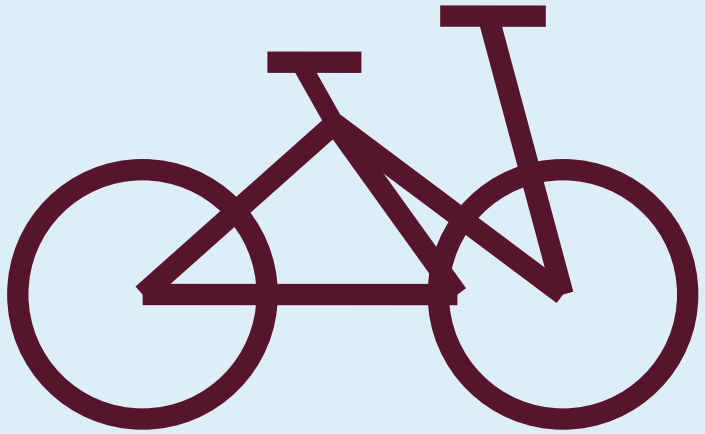
THE RHYTHM OF A CITY



Washington D.C. Bike Sharing

Scientific Data Visualization - 2026/27

Student: Di Vita Ivan

PROJECT ROADMAP

01 **GOAL**

What shapes general demand?

02 **DATA**

Rentals divided into two time scales

03 **CLEAN**

Inspect and validate



04 **SIMPLE PATTERNS**

Make patterns visible

05 **EXPLANATION**

Statistics + clustering

06 **PREDICTION**

Out-of-time ML

Two views of the same bike-sharing system

HOURLY DATA

17,379

hours recorded

TIME SPAN

2 years

From 2011 to 2012

DAILY DATA

731

calendar dates

TOTAL RENTALS

3.3 million

From 2011 to 2012

Key variables found in the tables

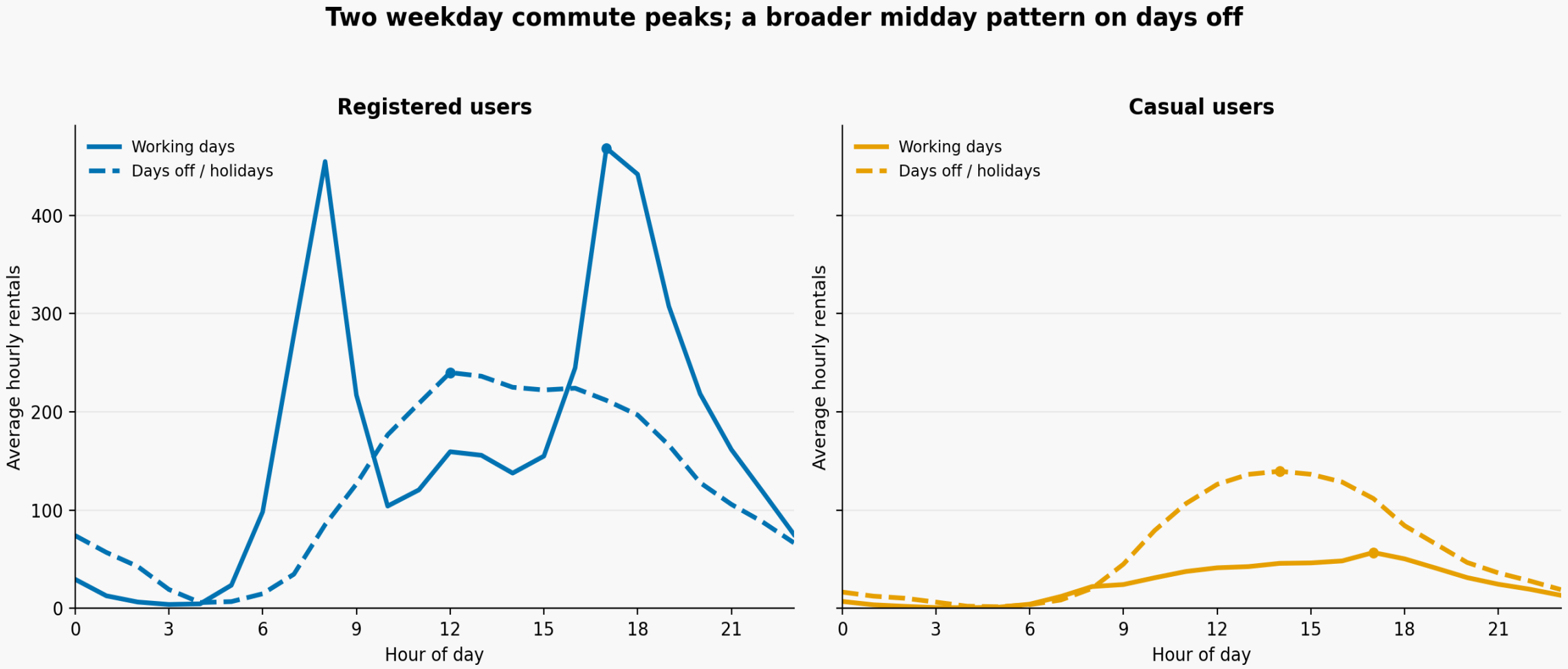
rentals (registered / casual), timestamp, working day, weather, temperature, humidity and wind

Data cleaning and data-quality inspection

01	**Inspect**	0 missing cells · 0 duplicate rows
02	**Convert**	dates, weekdays, temperature °C, humidity %
03	**Validate**	Check data integrity among the two tables.
04	**Optimize**	hourly gaps are not automatically zero rentals

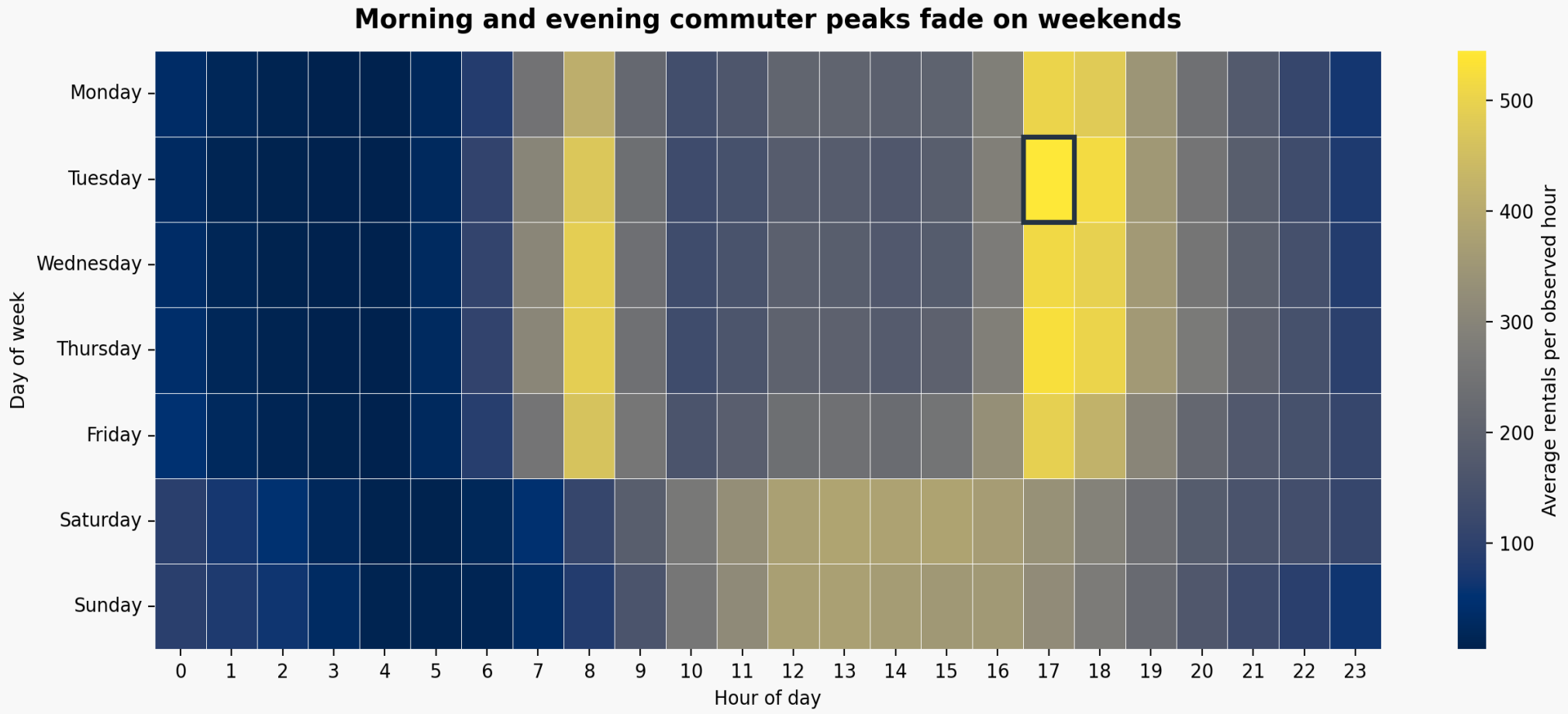
Some hourly slots are absent so complete-day analyses use 655 of 731 days.

User behavior



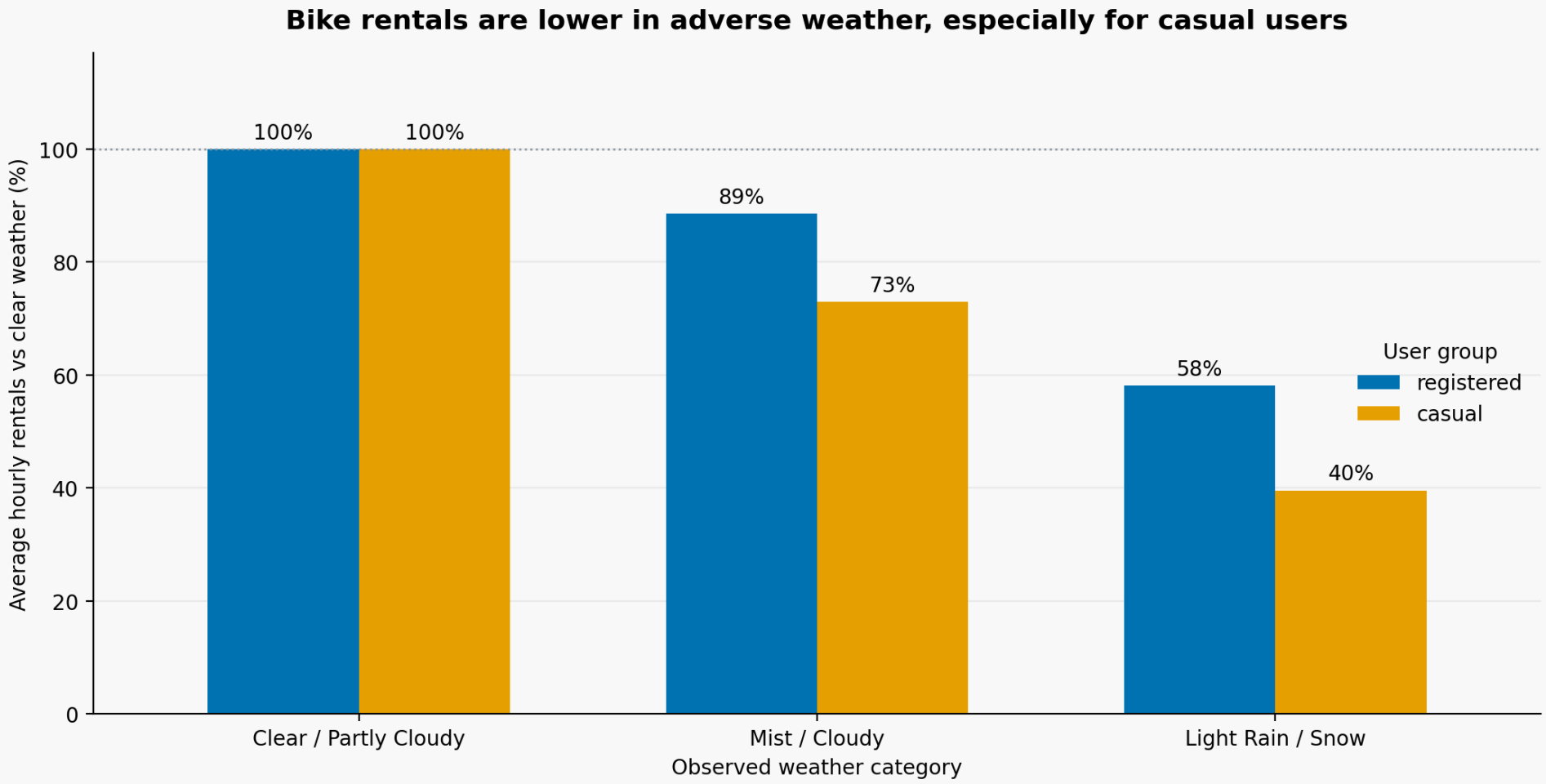
Registered riders peak near 08:00 and 17:00 (work hours) while casual riders rise toward midday.

Heatmap: weekly rhythm



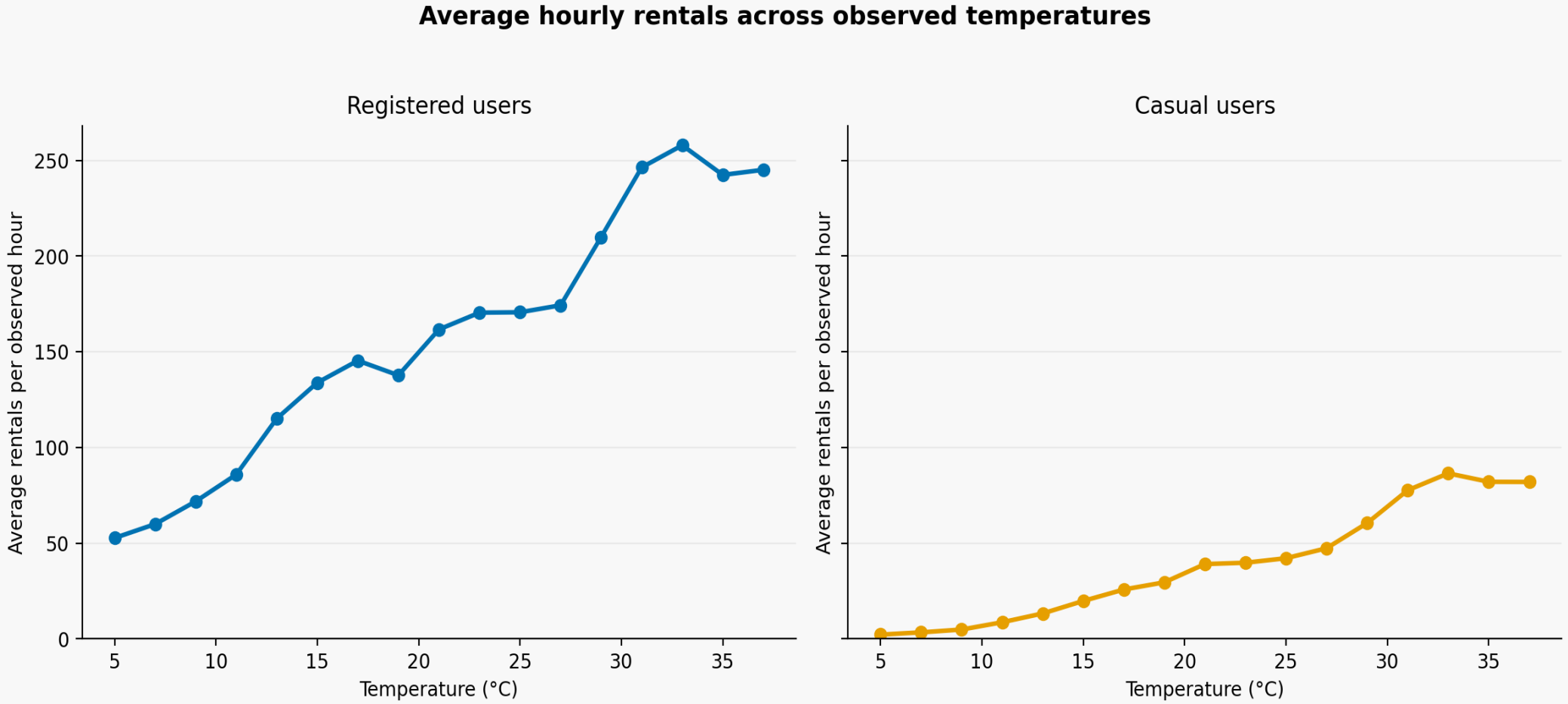
The heatmap (days x hours matrix) makes the workday–weekend contrast visible.

Adverse conditions coincide with lower bike usage



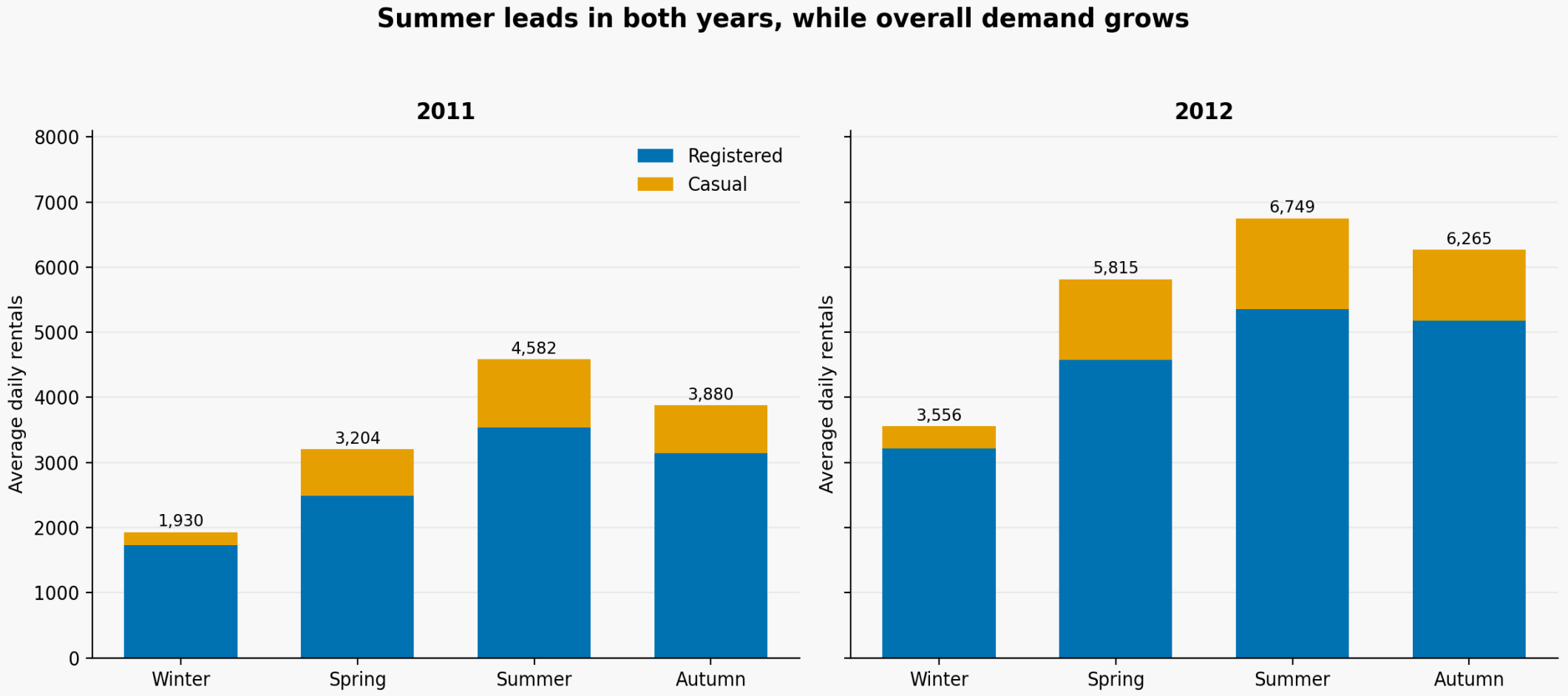
Light rain / snow: ~58% of clear-hour rentals for registered users; ~40% for casual.

Demand rises with temperature



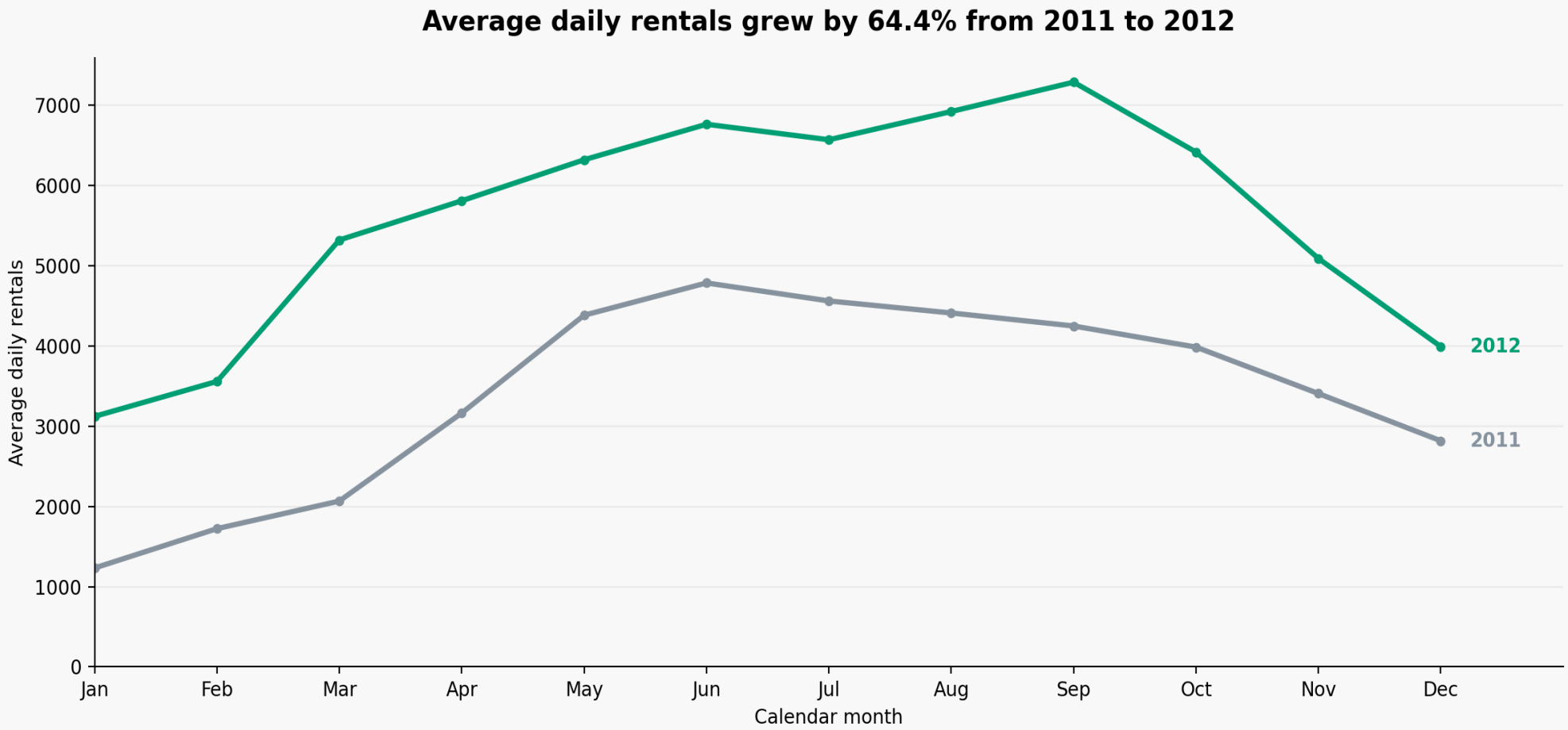
Hourly rentals tend to rise with the rise of temperature

Summer leads in both years and demand is growing



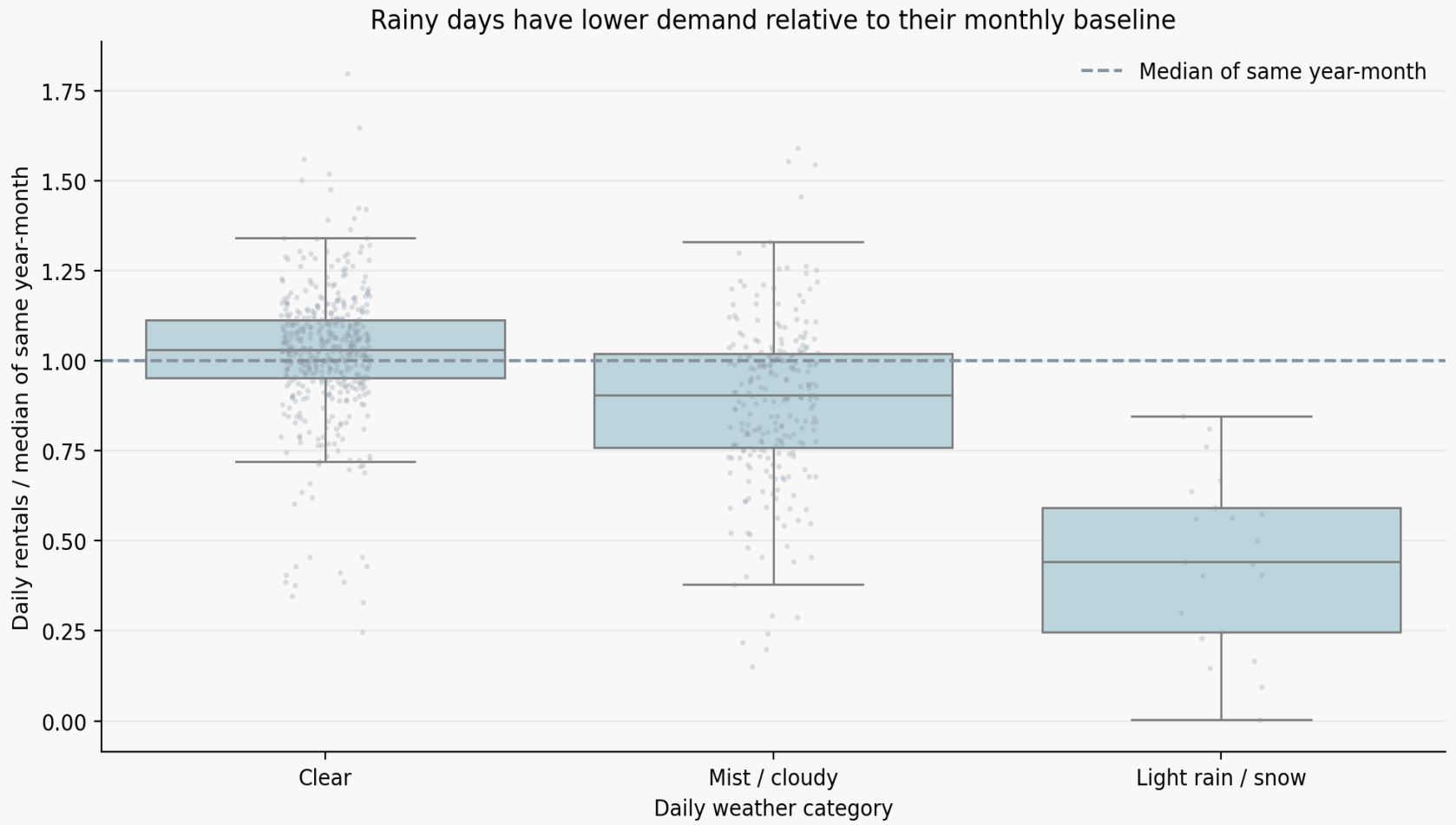
Summer is the peak season for bike rental.

Average daily rentals increased by 64.4% in 2012



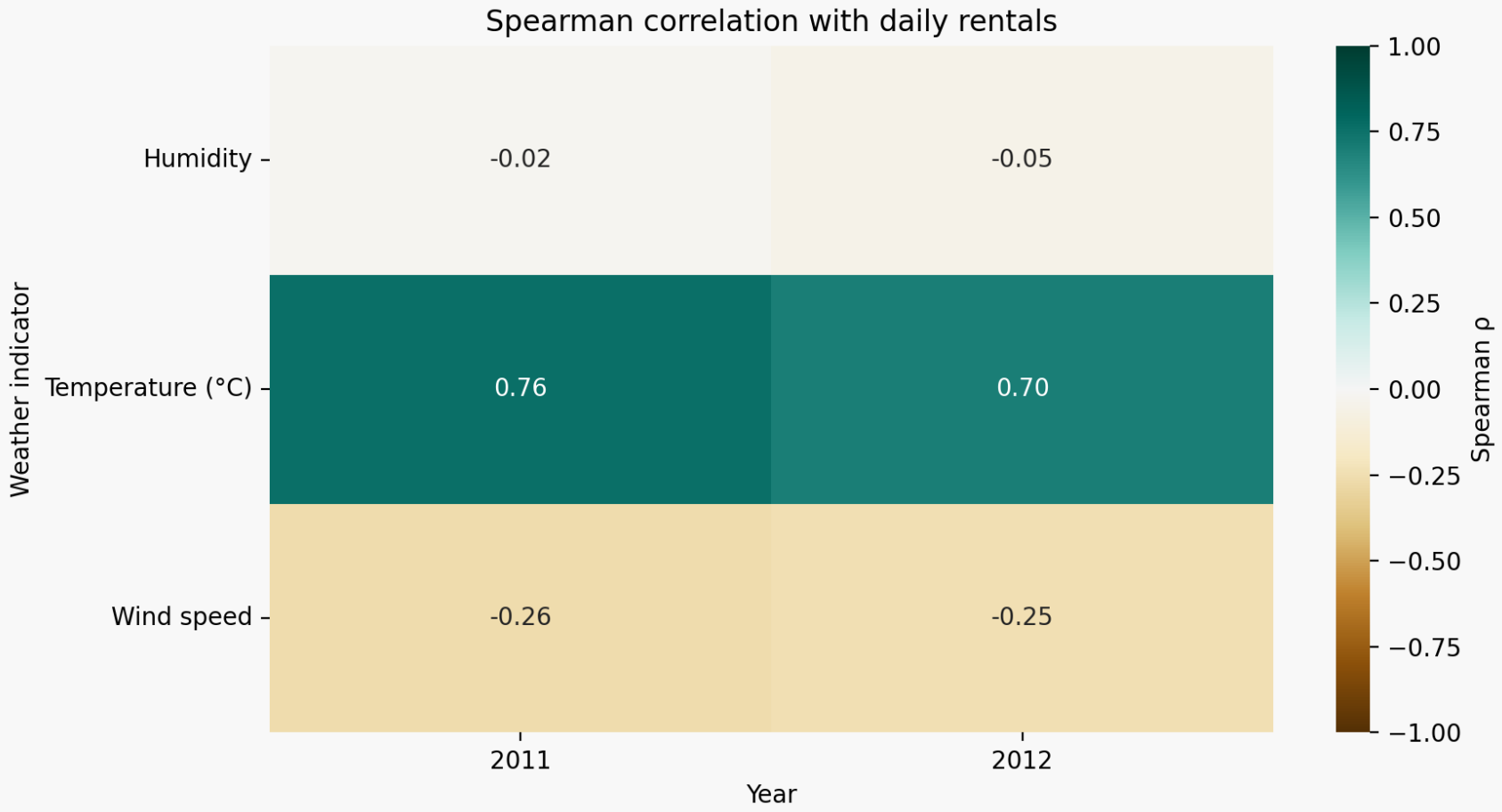
2012 has higher mean demand than 2011 comparing every month

Rainy days are far below their month’s demand baseline



Rainy days generally show lower demand relative to the monthly reference.

Temperature and demand rise together in both years



The temperature–demand association is positive in both years.

One day becomes a 24-dimensional profile

K-means sees the distribution of rentals across a day—not the calendar label.

01

731 days

All calendar dates

02

655 days

Complete 24-hour profiles

03

24 features

Hourly share of daily rentals (%)

04

K-means

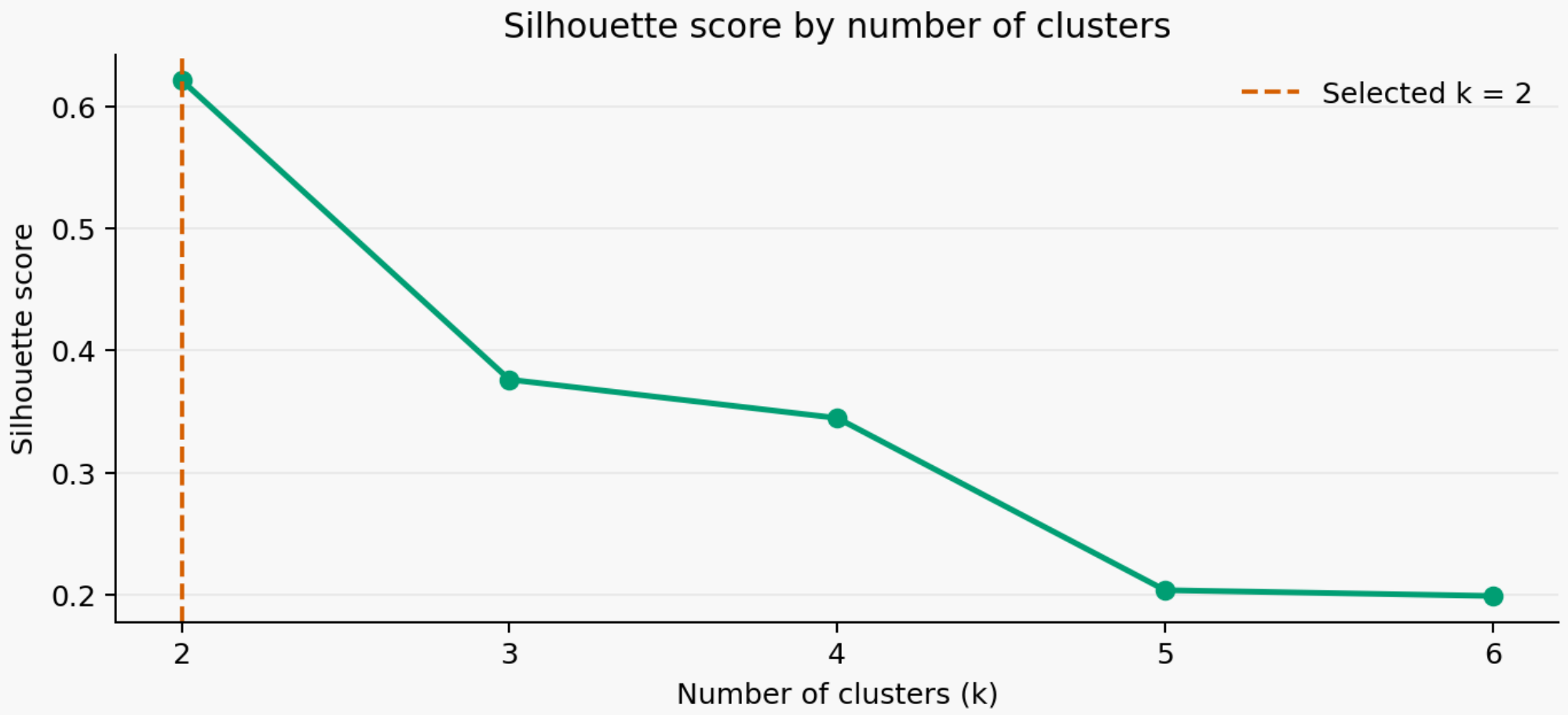
Group similar days

05

Explanation

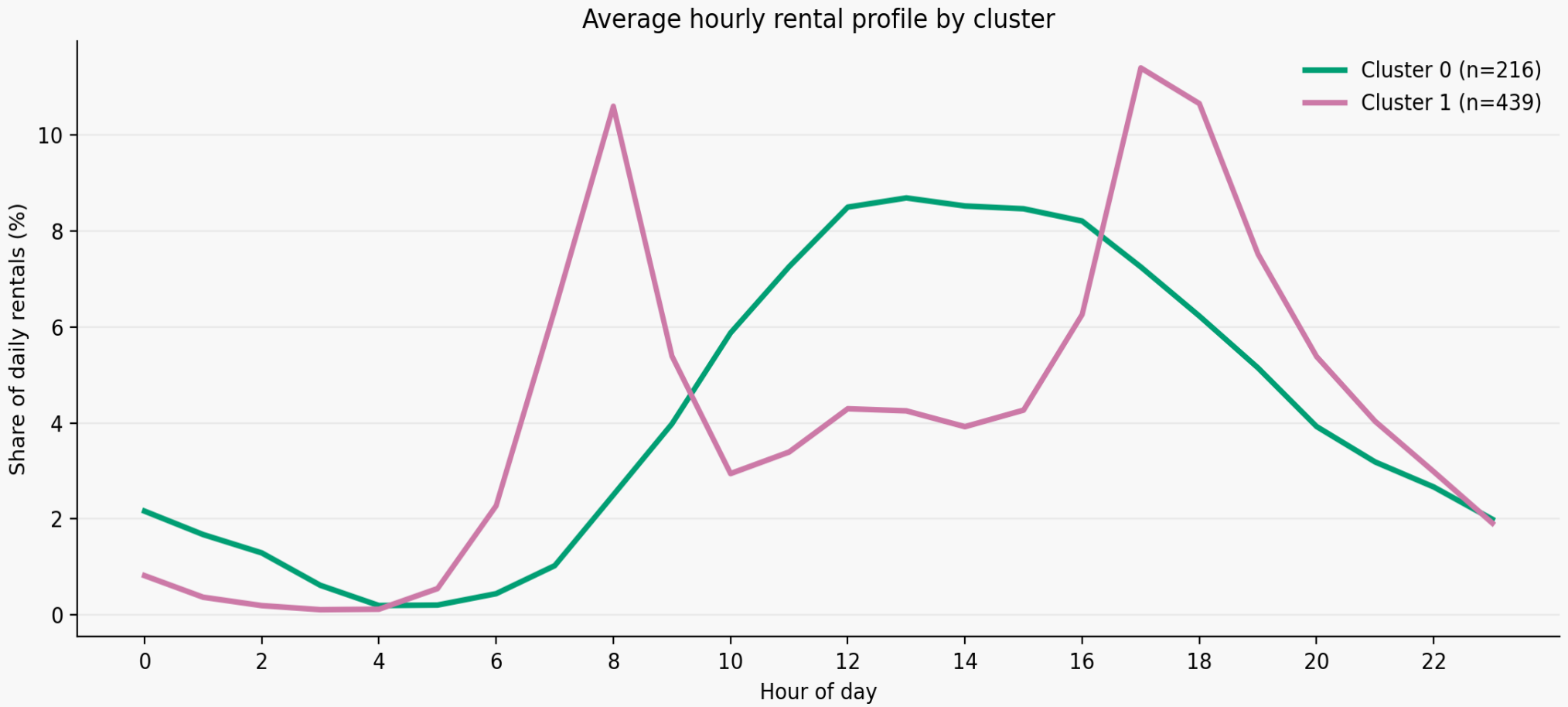
Interpret using calendar data

Two daily-pattern groups maximize silhouette



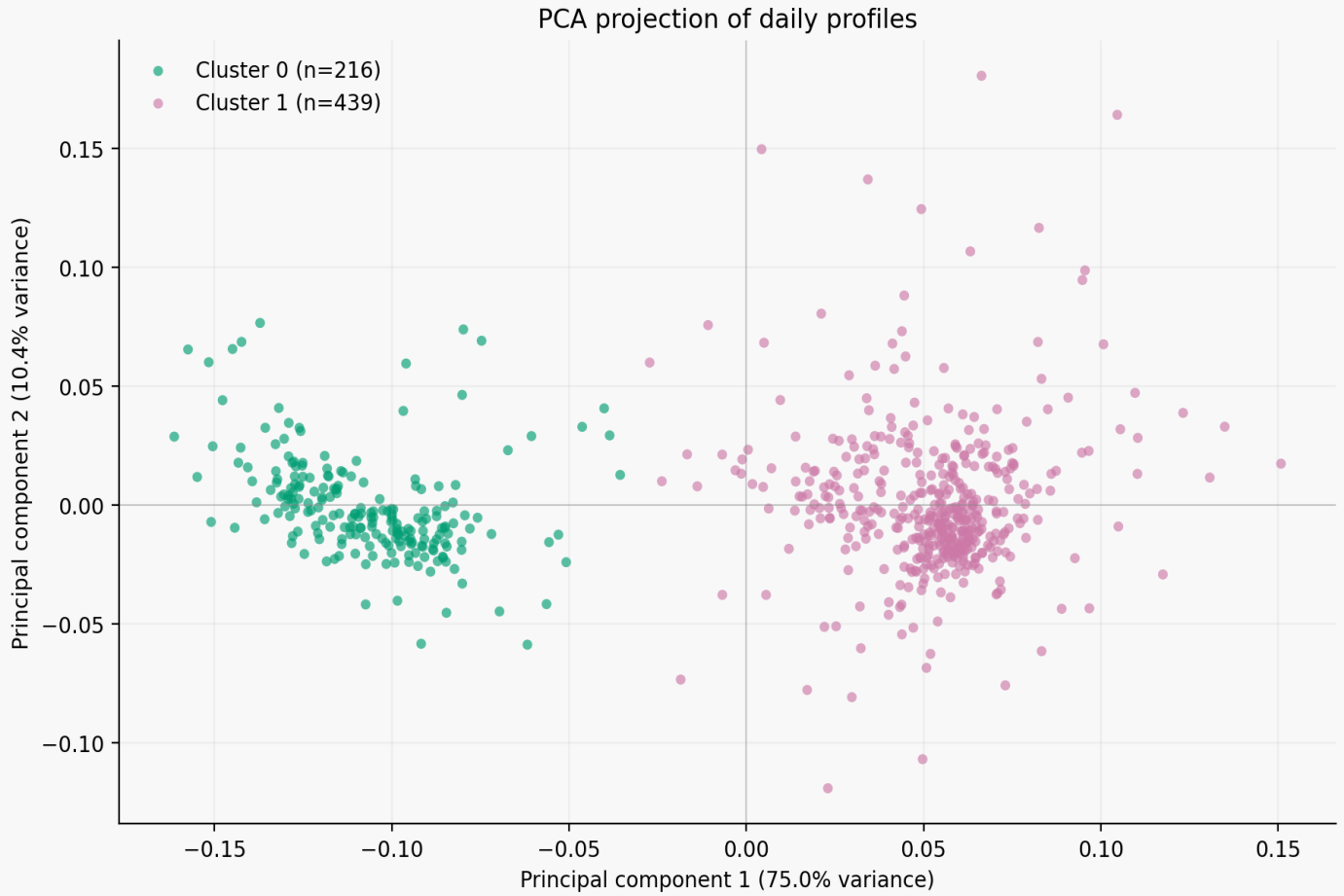
K = 2 has silhouette 0.621, higher than solutions with 3–6 groups.

Cluster: two interpretable rhythms emerge without labels



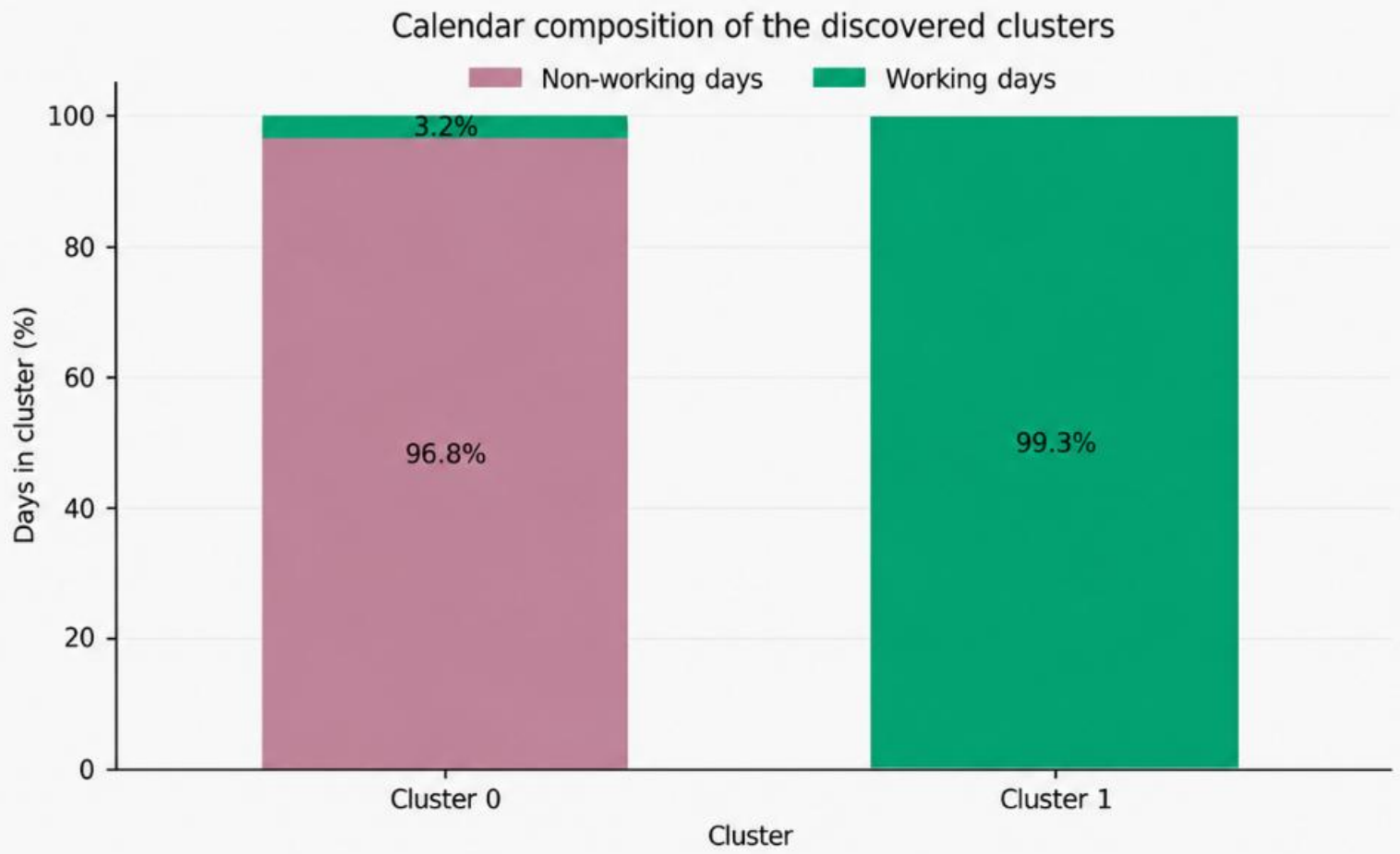
One profile peaks twice, the other rises broadly around midday.

PCA: Two components preserve 85.4% of profile variance



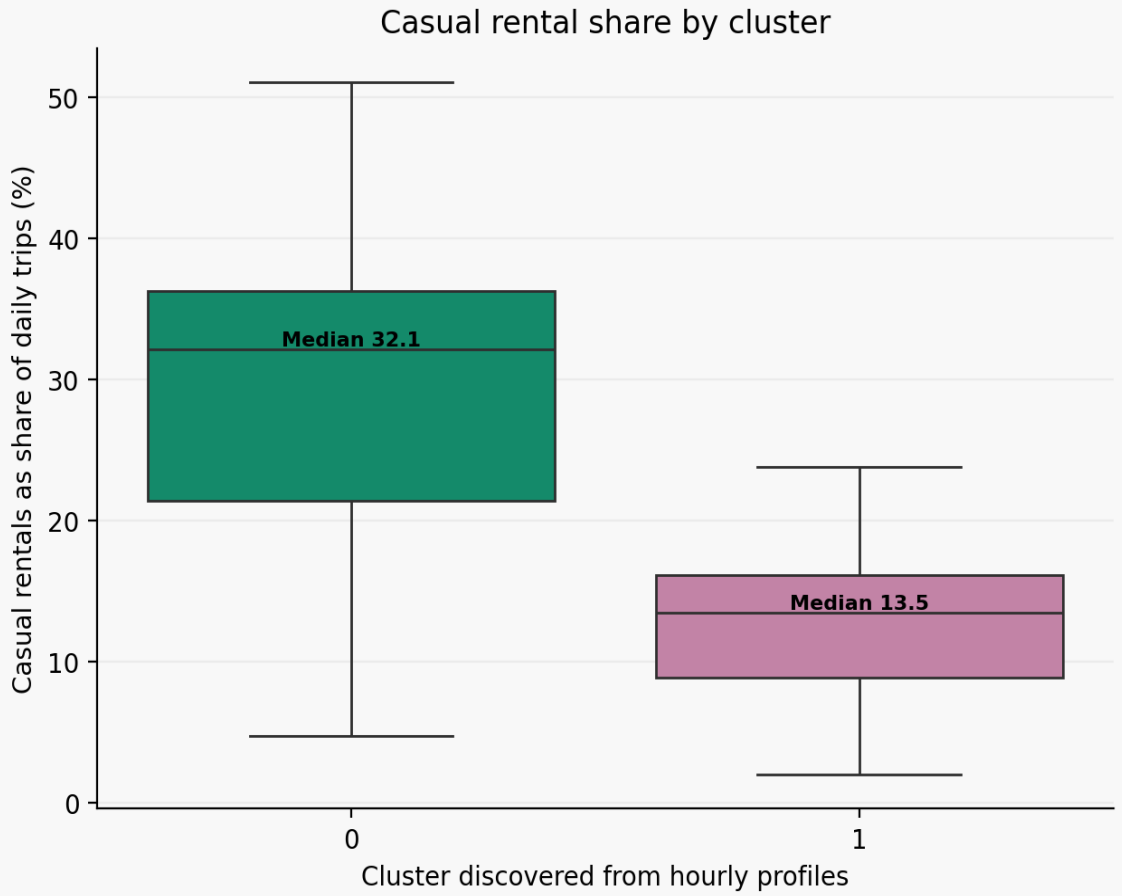
PC1 explains 75.0%; PC2 explains 10.4% of the 24-feature variation.

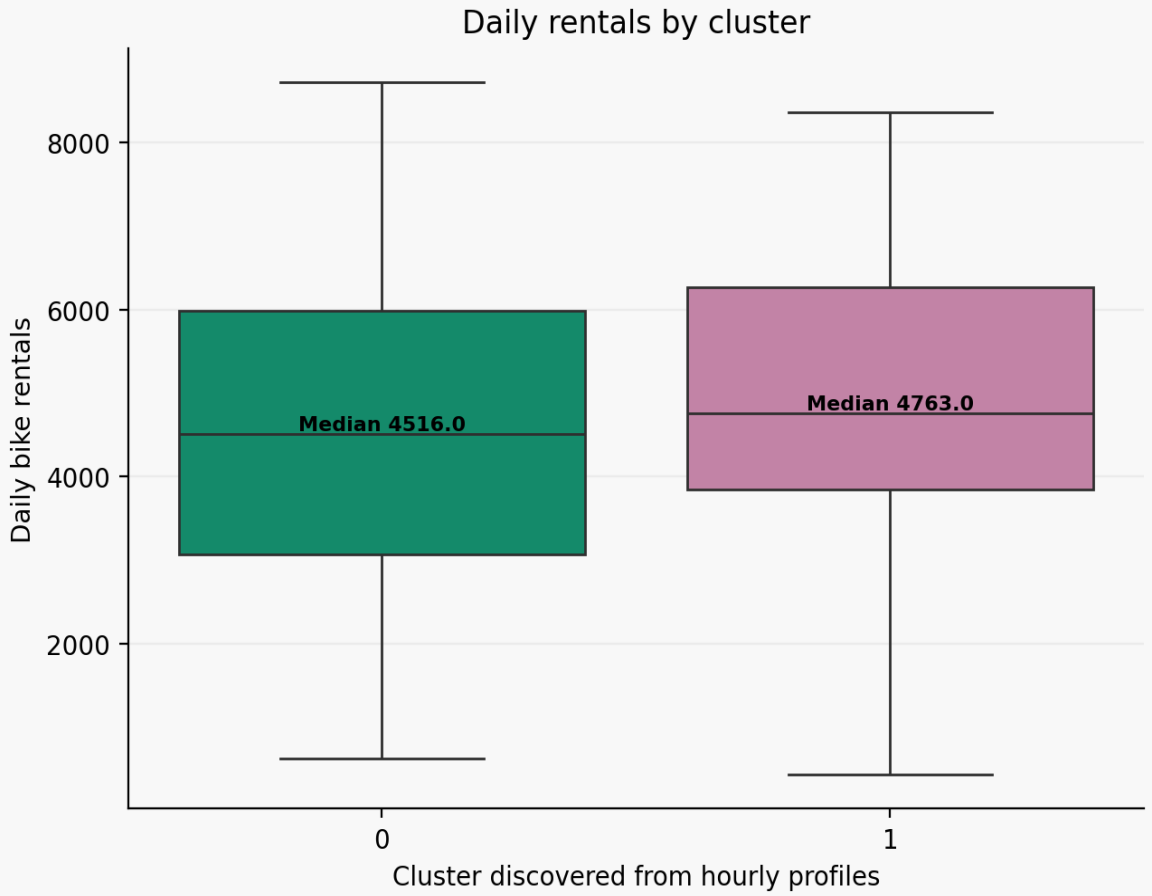
Calendar labels help interpret the discovered clusters



One group is 96.8% days off; the other 99.3% working days.

Clusters differ in who rides, not just when





Median casual share: 32.1% in days-off pattern vs 13.5% in commute pattern.

A supervised model must beat a simple baseline

How? Random Forest regression on weather and calendar information only

TRAINING HOURS

13,891

TEST PERIOD BEGINS

07 AUG 2012

HELD-OUT HOURS

3,488

RENTALS

3,292,679

Random Forest predicts hourly demand much better than baseline

HELD-OUT TEST PERFORMANCE

101.5

Baseline MAE · rentals/hour

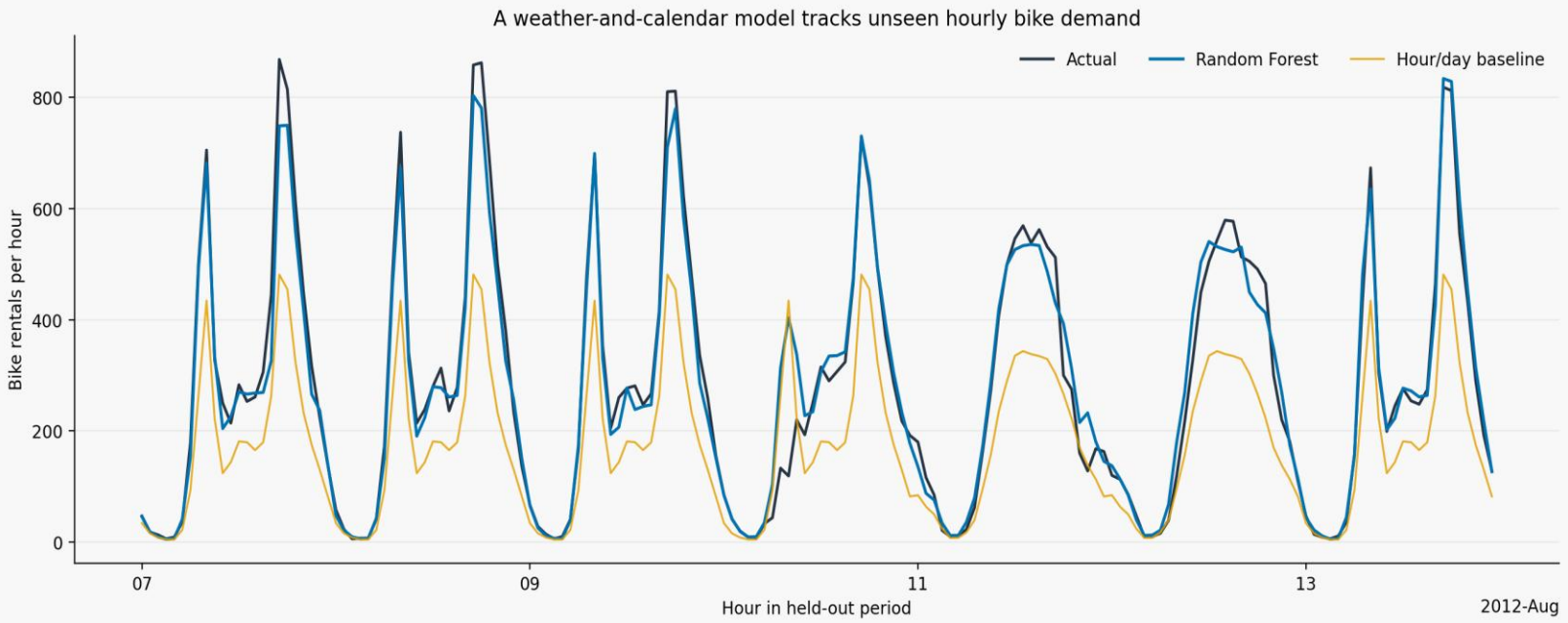
47.6

Random Forest MAE

53% lower error

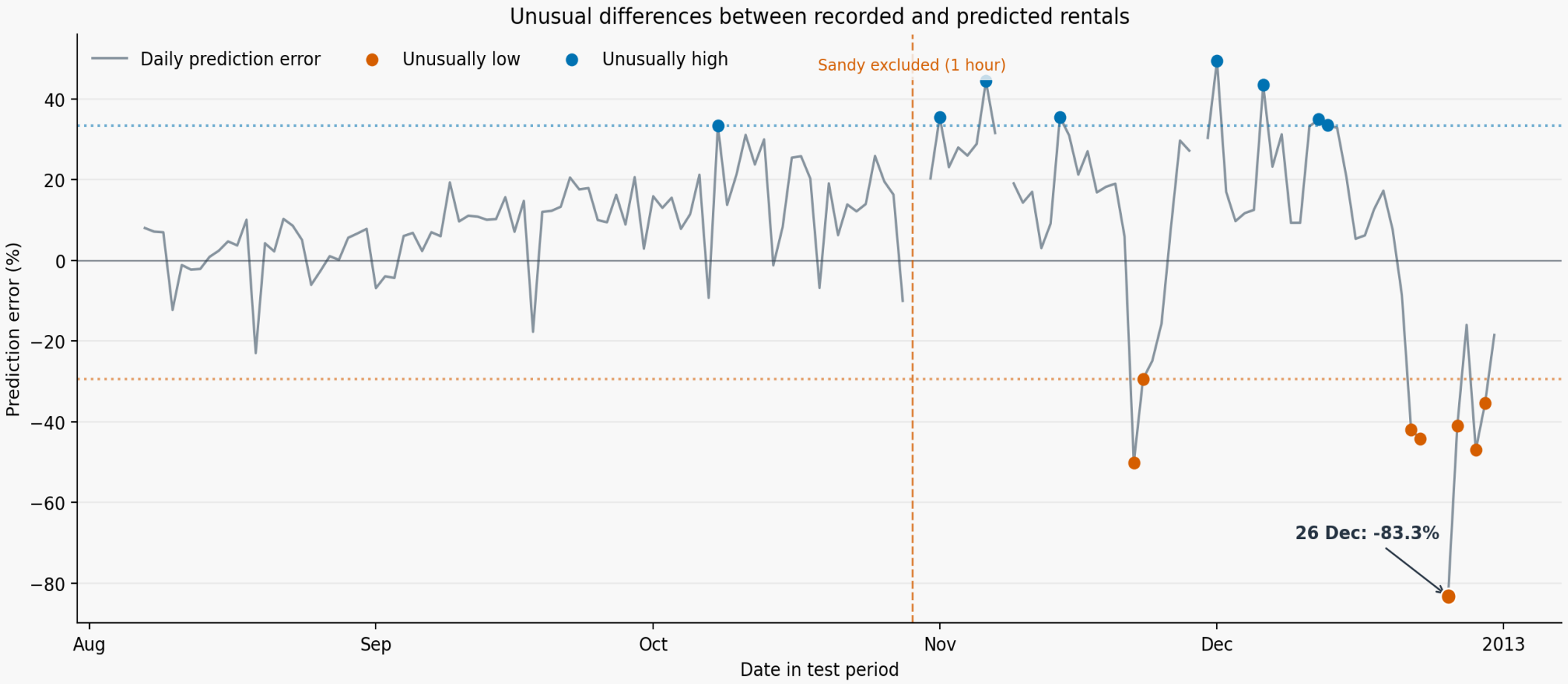
R^2 **0.57 → 0.89**

ACTUAL vs PREDICTED · 7–14 AUG 2012



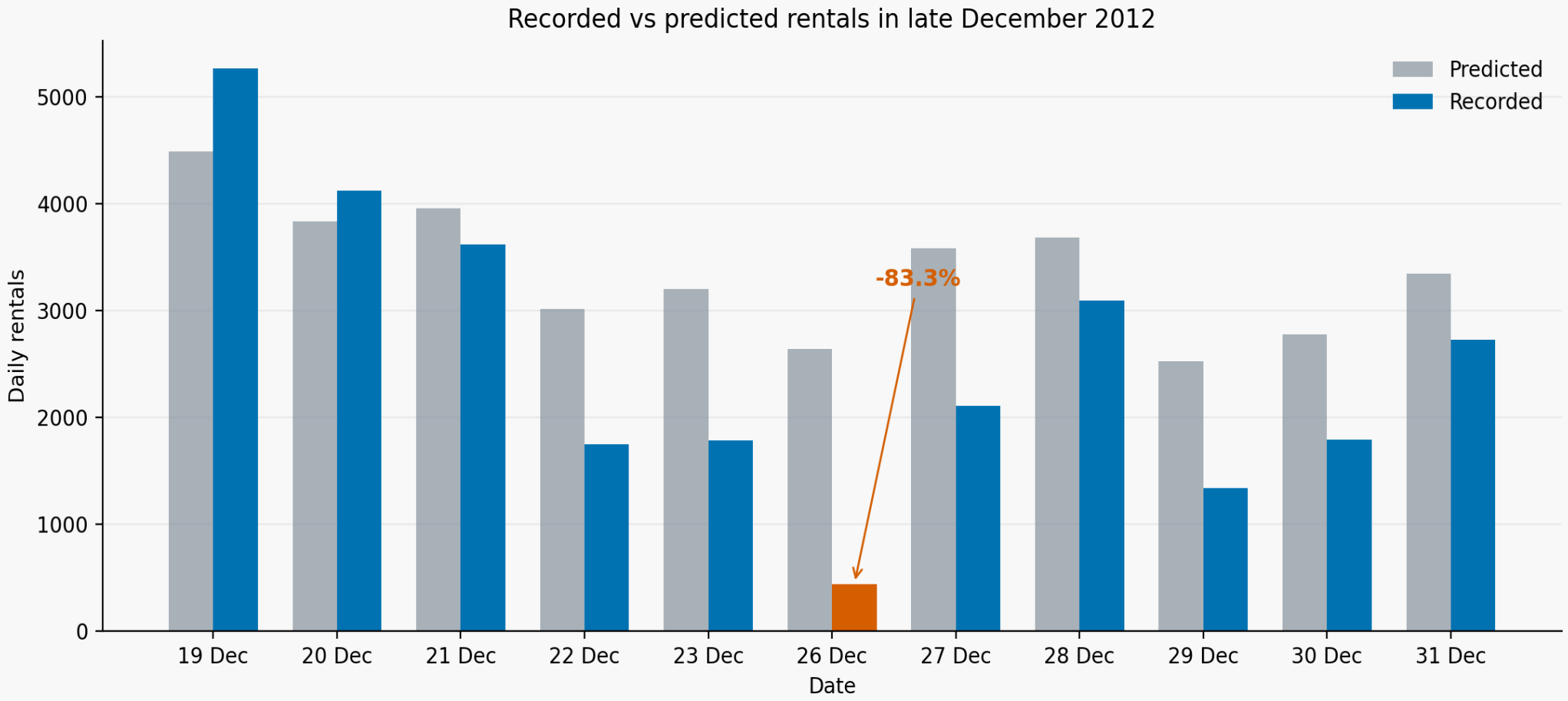
MAE is reduced with the model, and also the coefficent of determination got a better value

26 December 2012: an 83% gap between rentals and predictions



26 December has the largest negative prediction error in the test period

26 December 2012: observed demand was far below the model estimate



Observed demand was much lower than predicted during the Christmas period.

LIMITATIONS

OBSERVATIONAL

Weather associations do not demonstrate causal effects.

HISTORICAL DATA

Washington, D.C. only, from 2011 to 2012.

INCOMPLETE DAYS

76 days lack at least one recorded hour; Sandy has only one.

NO TRIP MOTIVES

Counts do not reveal purposes, routes or disruptions.

THANK YOU

FOR YOUR ATTENTION

DATA Fanaee-T & Gama (2013). Bike Sharing Dataset. UCI Repository. https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset
WEATHER Washington Reagan Airport, 26 Dec 2012: 1.42 in precipitation. NOAA-derived daily record via extremeweatherwatch.com/cities/washington-dc/year-2012

STUDENT Ivan Di Vita